

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

At the surface modality level, the benchmark and deployment match: both produce text outputs (binary classification labels, abstractive text summaries) [\[Q91, Q98\]](#). ROUGE [\[Q98–Q100\]](#) and the human rubric [\[Q106\]](#) are evaluation metrics that can in principle be adapted to Spanish reference summaries with appropriate tokenization. However, the benchmark provides no documentation of multilingual evaluation, and the human-written reference summaries [\[Q87\]](#) are English. UN OCHA situation reports follow specific genre conventions (numbered bullets, standardized headers, percentages) that differ from the abstractive narrative style elicited in the benchmark. The deployment also requires bilingual reporting (Spanish for national counterparts, English for global reporting), which the benchmark does not address. So the modality matches but the formatting/genre/language alignment does not.

ID	Evidence
Q87	"Since each timeline in our dataset is annotated by three workers (§3.2), it includes three summaries." (p.7)
Q91	"For evaluation, we use the average timeline-level accuracy." (p.7)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Strengths

- Text-based abstractive summarization is the right surface form for humanitarian SITREP production [\[Q98\]](#); the multi-reference ROUGE setup [\[Q100\]](#) is a reasonable template that extends to Spanish references with multilingual tokenization; the human evaluation protocol design [\[Q103–Q108\]](#) is methodologically sound and could be re-instantiated with bilingual evaluators.

ID	Evidence
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)
Q103	"We further conducted a human evaluation to better assess the quality of the summaries." (p.8)
Q104	"We take the whole TEST set and we evaluate eight summaries per timeline: 1) the three human-written summaries; 2) three model-generated summaries using the three models we develop for timeline summarization in the oracle setting; 3) two summaries from random systems: a random tweet that is part of the timeline, and a summary from a randomly selected timeline that belongs to a different crisis domain." (p.8)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q107	"Detailed instructions and annotation workflows are presented as Figures 15–21 in Appendix E." (p.8)

Q108	"Looking at the results over all the timelines, the human-written summaries are significantly better than the ones generated by models, in terms of overall quality with an average rating of 4.12." (p.8)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Text-output modality matches deployment needs (analyst-consumable text summaries and binary relevance flags) [Q91, Q98].

OF-2: Text-to-speech is not a primary requirement — analysts are literate professionals (region YAML literacy_and_education.analyst_population). Not applicable.

OF-3: Analyst literacy is high in Spanish and English [region YAML]; no accessibility barrier at the consumer side. Affected populations have ~94–96% national literacy [WEB-2, WEB-3] but with rural and indigenous gaps [WEB-5] — relevant if outputs were redirected to community use, but not for the primary analyst use case.

OF-4: External validity is partially preserved at the modality level but compromised by language (English-only references [Q87], English-anchored Coherence [Q153]) and genre (no SITREP conventions). ROUGE applied to English references against Spanish outputs would be invalid.

ID	Evidence
Q87	"Since each timeline in our dataset is annotated by three workers (§3.2), it includes three summaries." (p.7)
Q91	"For evaluation, we use the average timeline-level accuracy." (p.7)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-2	Ecuador adult literacy ~94% (2022). (https://www.theglobaleconomy.com/Ecuador/literacy_rate/)
WEB-3	Ecuador adult literacy ~96.3% (2022). (https://www.worldviewdata.com/countries/ecuador/literacy-rate)
WEB-5	Indigenous communities have lower literacy rates than national average. (https://iwgia.org/en/ecuador/5087-iw-2023-ecuador.html)

Information Gaps

- Whether multilingual ROUGE/BERTScore implementations adopted by humanitarian NLP practitioners are validated for Spanish SITREP-style text.

Requires Expert Verification

- Specific UN OCHA / Cruz Roja SITREP genre conventions to anchor evaluation rubric adaptation.

Remediation

Gap 1: ROUGE and the human rubric assume English references; no Spanish or bilingual evaluation; no SITREP genre conventions.

Recommendation: Use multilingual semantic-similarity metrics (e.g., multilingual BERTScore) against Spanish reference summaries; develop a parallel English/Spanish evaluation track aligned with UN OCHA SITREP and Cruz Roja appeal genre conventions.